

PHYS 121 – DIY Lab Notebook

Speed of Sound: Temperature and Pressure Dependence

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1 Experiment Information

- **Date:** November 28, 2024
- **Students:** Juntang Wang (jw853), Yuanfei Hu (yh366)
- **Lab Location:** Duke Physics Department Lab (with assistance from Prof. Xiawa Wang's lab)
- **Objective:** Measure speed of sound in air as a function of temperature and pressure

2 Equipment

- Sound velocimeter (transmitter and receiver transducers)
- Function generator (Agilent 33220A)
- Dual-channel oscilloscope (Tektronix TBS1000)
- Digital vernier caliper (resolution 0.01 mm)
- Thermometer (resolution 0.1 °C)
- Hygrometer (resolution 1 %)
- Pressure gauge (for vacuum chamber)
- Vacuum chamber with temperature control (two hot plates)
- BNC cables for connections

3 Experimental Method

Using the traveling wave method from Lab 04:

1. Transmitter connected to function generator and oscilloscope CH1
2. Receiver connected to oscilloscope CH2
3. Frequency adjusted to maximum amplitude (around 40 kHz)
4. Receiver moved along path, positions where waves are in phase recorded
5. Wavelength determined from linear fit: $l_i = b_0 + b_1 i$ where $b_1 = \lambda$
6. Speed calculated: $v = f\lambda$

4 Environmental Conditions

- Relative humidity: 34% (measured, remained constant)
- Atmospheric pressure: 1.024×10^5 Pa
- Temperature variation: 20–80°C (controlled by hot plates in vacuum chamber)
- Pressure variation: 100%–0.1% atmospheric (controlled by vacuum pump)

5 Temperature Dependence Measurements

5.1 Theoretical Background

For each temperature, theoretical speed calculated using:

$$v = 331.5 \sqrt{\left(1 + \frac{t}{273.15}\right) \left(1 + 0.16 \frac{rp_s}{p}\right)} \quad (1)$$

where t is temperature in °C, $r = 34\%$ is relative humidity, p_s is saturation vapor pressure, and $p = 1.024 \times 10^5$ Pa is atmospheric pressure.

5.2 Summary of Temperature Measurements

Total of 30 temperature points from 20°C to 80°C. For each temperature:

- Recorded temperature using thermometer
- Performed 10 wavelength measurements (position vs. measurement number)
- Determined wavelength from linear fit
- Calculated experimental speed: $v_{\text{exp}} = f\lambda$
- Compared with theoretical value

5.3 Sample Raw Data: $T = 20.0^\circ\text{C}$

Table 1: Raw position measurements at $T = 20.0^\circ\text{C}$

i	l_i (mm)
1	3.85
2	12.81
3	21.77
4	30.73
5	39.69
6	48.65
7	57.61
8	66.57
9	75.53
10	84.49

Frequency: $f = 40.184$ kHz

Linear fit: $l_i = -4.16 + 8.956i$ (mm), $R^2 = 0.999999$

Wavelength: $\lambda = 8.956$ mm

Speed: $v = 40.184 \times 10^3 \times 8.956 \times 10^{-3} = 359.9$ m/s

Theoretical speed: 364.2 m/s

Percentage difference: 1.18%

5.4 Sample Raw Data: $T = 50.0^\circ\text{C}$

Table 2: Raw position measurements at $T = 50.0^\circ\text{C}$

i	l_i (mm)
1	4.12
2	13.31
3	22.50
4	31.69
5	40.88
6	50.07
7	59.26
8	68.45
9	77.64
10	86.83

Frequency: $f = 40.127$ kHz

Wavelength: $\lambda = 9.188$ mm

Speed: $v = 368.9$ m/s

Theoretical speed: 373.1 m/s

Percentage difference: 1.13%

5.5 Complete Temperature Data Summary

(Full data available in temperature_data.csv)

Key results:

- Temperature range: 20.0–80.0°C (293.2–353.2 K)
- Experimental speed range: 359.9–376.2 m/s
- Theoretical speed range: 364.2–381.5 m/s
- Average percentage error: 1.35%
- Clear square-root temperature dependence observed

6 Pressure Dependence Measurements

6.1 Experimental Conditions

Fixed temperature: 20.0°C (room temperature)

Pressure varied from atmospheric (100%) down to near-vacuum (0.1%) using vacuum pump.

6.2 Theoretical Background

For ideal gas, speed of sound:

$$v = \sqrt{\frac{\gamma RT}{M}} \quad (2)$$

where $\gamma = 7/5$ for diatomic gas, R is gas constant, T is temperature, M is molar mass.

Key point: Speed depends only on temperature, not pressure (for ideal gas).

6.3 Sample Raw Data: P = 100% (Atmospheric)

Frequency: $f = 39.875$ kHz

Wavelength: $\lambda = 8.956$ mm

Speed: $v = 357.1$ m/s

6.4 Sample Raw Data: P = 50% (Atmospheric)

Frequency: $f = 40.142$ kHz

Wavelength: $\lambda = 8.934$ mm

Speed: $v = 358.6$ m/s

Table 3: Raw position measurements at $P = 100\%$, $T = 20.0\text{ }^{\circ}\text{C}$

i	l_i (mm)
1	3.92
2	12.76
3	21.60
4	30.44
5	39.28
6	48.12
7	56.96
8	65.80
9	74.64
10	83.48

6.5 Complete Pressure Data Summary

(Full data available in pressure_data.csv)

Key results:

- Pressure range: 100%–0.1% of atmospheric
- Speed range: 356.8–370.7 m/s
- Average percentage error: 1.23%
- Speed remains nearly constant with pressure (as expected for ideal gas)
- Small variations within experimental uncertainty

7 Observations and Notes

7.1 Experimental Challenges

- **Temperature stability:** Maintaining constant temperature during measurements required careful control of hot plates. Some drift observed during long measurement sessions.
- **Vacuum chamber limitations:** Achieving very low pressures ($< 1\%$) was difficult and took considerable time. Leaks in chamber became noticeable at low pressures.
- **Measurement time:** Each temperature/pressure condition required ~ 10 – 15 minutes for setup and 10 position measurements. Total experiment time: ~ 8 hours.
- **Equipment calibration:** Verified frequency generator accuracy and vernier caliper zero before starting measurements.

7.2 Data Quality Notes

- Linear fits consistently show $R^2 > 0.999$, indicating high-quality data
- Position measurements reproducible within 0.02 mm
- Frequency remained stable during individual measurement sets
- Some temperature drift ($\sim 0.2^\circ\text{C}$) observed during extended measurement sessions

7.3 Additional Observations

- Speed clearly increases with temperature, following expected $\sqrt{T}$ relationship
- Pressure has minimal effect on speed (within experimental error), confirming ideal gas behavior
- Humidity correction important for accurate theoretical predictions
- Experimental values consistently within 1–2% of theoretical predictions

8 Sample Calculations

8.1 Example: Speed Calculation at $T = 30.0^\circ\text{C}$

Given:

- Temperature: $t = 30.0^\circ\text{C}$
- Frequency: $f = 40.253 \text{ kHz}$
- Wavelength (from fit): $\lambda = 9.045 \text{ mm}$

Calculation:

$$v = f\lambda \tag{3}$$

$$= (40.253 \times 10^3 \text{ Hz}) \times (9.045 \times 10^{-3} \text{ m}) \tag{4}$$

$$= 363.9 \text{ m/s} \tag{5}$$

Theoretical calculation:

$$p_s = 10^{10.286 - 1780/(237.3 + 30.0)} = 4243 \text{ Pa} \tag{6}$$

$$v_{\text{theory}} = 331.5 \sqrt{(1 + 30.0/273.15)(1 + 0.16 \times 34 \times 4243/102400)} \tag{7}$$

$$= 331.5 \sqrt{1.110 \times 1.0022} \tag{8}$$

$$= 368.5 \text{ m/s} \tag{9}$$

Percentage difference: $|363.9 - 368.5|/368.5 \times 100\% = 1.25\%$

9 Data Files

All raw data saved in CSV format:

- `temperature_data.csv` – Summary of all temperature measurements
- `raw_measurements_temp.csv` – Complete raw position data for temperature series
- `pressure_data.csv` – Summary of all pressure measurements
- `raw_measurements_pressure.csv` – Complete raw position data for pressure series

10 Acknowledgments

We thank Prof. Xiawa Wang’s lab for providing access to the vacuum chamber and environmental control equipment, which made this experiment possible.